KAUSHIK ROY

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ACADEMIC APPOINTMENTS

• 2025–present: Assistant Professor, Department of Computer Science, The University of Alabama.

EDUCATION

• Ph.D., Computer Science and Engineering, University of South Carolina, 2025. Dissertation: Process-Grounded Knowledge-Infused Learning and Decision Making. Advisor: Amit Sheth.

• M.S., Computer Science, Indiana University Bloomington, 2017.

• B.E., Computer Science, R.V. College of Engineering, 2015.

RESEARCH INTERESTS

Neurosymbolic and knowledge-infused AI; process-grounded learning; causal reasoning; trustworthy and agentic AI; sequential decision-making; healthcare and manufacturing applications.

RESEARCH APPOINTMENTS AND AFFILIATIONS

• BRAINS Lab, The University of Alabama — research affiliation.

• Artificial Intelligence Institute, University of South Carolina — Graduate Research Assistant, 2020–2025.

• Bosch Center for Artificial Intelligence — AI Researcher, summer 2024 and summer 2023.

• Samsung Research America — AI Researcher, summer 2022.

• The Alan Turing Institute — Visiting Researcher, fall 2021.

• StARLinG Lab, University of Texas at Dallas — Research Assistant, 2017–2020.

RESEARCH SUPPORT AND PROPOSAL ACTIVITY

• 2026 UA SCoE CORE Equipment Funding ($30,000 multi-investigator award) — listed collaborator; individual allocation not asserted.

• Contributor to NSF EAGER #2335967, Knowledge-Guided Neurosymbolic AI with Guardrails for Safe Virtual Health Assistants (PI: Amit Sheth).

• Contributor to NSF EAGER #2133842, Advancing Neurosymbolic AI with Deep Knowledge-Infused Learning (PI: Amit Sheth).

• 2026: Research proposal development in trustworthy neurosymbolic AI, manufacturing datasets, and causal decision support; submissions are not represented as awards.

PEER-REVIEWED PUBLICATIONS AND SCHOLARLY OUTPUTS

The following is a consolidated record of newer publications and the bibliography from the author's earlier public academic CV. Publication metadata should be checked against publisher records for formal applications.

• Roy, K., et al. (2026). Exploring the Potential of Large Language Models for Assisting with Mental Health Diagnostic Assessments. ACM Transactions on Computing for Healthcare.

• Ramnani, S., Roy, K., and Sheth, A. (2026). MATRIX: Mental heAlth diagnostics Through Real time Intelligent unified X-AI attribution reasoning. Frontiers in Digital Health.

• Chattopadhyay, A., Dandekar, R., and Roy, K. (2026). Metatuning: An Empirical Study of Judge-Guided Prompt Refinement and Its Boundary Conditions. Neurosymbolic Artificial Intelligence.

• Tripathi, H., Roy, K., Neupane, S., and Rahimi, S. (2026). How Compliant is Sepsis Treatment? An Expert-Guided Neuro-symbolic Pipeline for Generating Clinical Compliance Insights. NeSy 2026.

• Garimella, R., Roy, K., et al. (2026). In-Situ Eval: A Modular Framework for Custom and Real-Time RAG Benchmarking. AAAI Demonstration Track.

BOOK CHAPTERS

• Kaushik Roy, Wu, S., Oltramari, M. (2025). "Neurosymbolic Cognitive Methods for Enhancing Foundation Model-based Reasoning." (to appear in: The Handbook on Neurosymbolic AI and Knowledge Graphs, IOS Press) [chapter preprint].

• Kaushik Roy, Gaur, M., Soltani, M., Rawte, V., Kalyan, A., and Sheth, A. (2024). "Proknow: Process Knowledge for Safety Constrained and Explainable Question Generation for Mental Health Diagnostic Assistance in the Age of Large Language Models." Science and Technology - Recent Updates and Future Prospects (to appear in Vol. 2) [chapter preprint].

JOURNAL PUBLICATIONS

• Venkataramanan, R., Tripathy, A., Kumar, T., Serebryakov, S., Justine, A., Shah, A., Bhattacharya, S., Foltin, M., Faraboschi, P., Kaushik Roy, and Sheth, A., Constructing a Metadata Knowledge Graph as an Atlas for Demystifying AI Pipeline Optimization (2025). Frontiers in Big Data (Vol. 7) [paper].

• Sheth, Amit, Kaushik Roy, Hemant Purohit, and Amitava Das (2024). "Civilizing and Humanizing Artificial Intelligence in the Age of Large Language Models." IEEE Internet Computing (Vol. 28, Issue 5) [paper].

• Kaushik Roy. Healthcare Assistance Challenges-Driven Neurosymbolic AI (2024). Biomedical Journal of Science & Technical Research (Vol. 58, Issue 2) [paper].

• Shyalika, C., Kaushik Roy, et al. (2024). "RI2AP: Robust and Interpretable 2D Anomaly Prediction in Assembly Pipelines." Sensors (Vol. 24, Issue 10) [paper].

• Sheth, A., and Kaushik Roy. (2024). "Neurosymbolic Value-Inspired Artificial Intelligence (Why, What, and How)." IEEE Intelligent Systems (Vol. 39, Issue 1) [paper].

• Sheth, A., Kaushik Roy, and Gaur, M. (2023). "Neurosymbolic Artificial Intelligence (Why, What, and How)." IEEE Intelligent Systems (Vol. 38, Issue 3) [paper].

• Kaushik Roy, Gaur, M., Soltani, M., Rawte, V., Kalyan, A., and Sheth, A. (2023). "Proknow: Process Knowledge for Safety Constrained and Explainable Question Generation for Mental Health Diagnostic Assistance." Frontiers in Big Data (Vol. 5) [paper].

• Chaudhuri, S.K., Kleppinger, J.W., Nag, R., Kaushik Roy, et al. (2021). "A CdZnTeSe Gamma Spectrometer Trained by Deep Convolutional Neural Network for radioisotope identification." In Hard X-Ray, Gamma-Ray, and Neutron Detector Physics XXIII (Vol. 11838) [paper].

• Sheth, A., Gaur, M., Kaushik Roy, Venkataraman, R., and Khandelwal, V. (2022). "Process Knowledge-Infused AI: Toward User-Level Explainability, Interpretability, and Safety." IEEE Internet Computing (Vol. 26, Issue 5) [paper].

• Sheth, A., Gaur, M., Kaushik Roy, and Faldu, K. (2021). "Knowledge-intensive Language Understanding for Explainable AI." IEEE Internet Computing (Vol. 25, Issue 5) [paper].

CONFERENCE PUBLICATIONS

• Tiwari, A., Sinan, M., Kaushik Roy, Sheth, A., Saha, S., & Bhattacharyya, P. (2024, August). Towards a Contextualized and Semantics Infused Dialogue Generation Loss Function and Evaluation Metric. In Joint European Conference on Machine Learning and Knowledge Discovery in Databases (pp. 342-360). Cham: Springer Nature Switzerland [paper].

• Pallagani, V., Kaushik Roy, et al. (2024). "On the Prospects of Incorporating Large Language Models in Automated Planning and Scheduling." In Proceedings of the International Conference on Automated Planning and Scheduling (Vol. 34, pp. 432-444) [paper].

• Kaushik Roy, Khandelwal, V., Vera, V., Surana, H., Heckman, H., & Sheth, A. (2024, March). GEAR-Up: Generative AI and External Knowledge-Based Retrieval: Upgrading Scholarly Article Searches for Systematic Reviews. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 38, No. 21, pp. 23823-23825) [paper].

• Venkataramanan, R., Kaushik Roy, Raj, K., Prasad, R., Zi, Y., Narayanan, V., & Sheth, A. (2023, October). Cook-gen: Robust Generative Modeling of Cooking Actions from Recipes. In 2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC) (pp. 981-986) [paper].

• Kaushik Roy, Garg, T., Palit, V., Zi, Y., Narayanan, V., and Sheth, A. (2023, June). Knowledge Graph Guided Semantic Evaluation of Language Models for User Trust. In 2023 IEEE Conference on Artificial Intelligence (CAI) (pp. 234-236) [paper].

• Kaushik Roy, Khandelwal, V., Goswami, R., Dolbir, N., Malekar, J., & Sheth, A. (2023, September). Demo alleviate: Demonstrating Artificial Intelligence Enabled Virtual Assistance for Telehealth: The Mental Health Case. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 37, No. 13, pp. 16479-16481) [paper].

• Gaur, M., Kaushik Roy, Sharma, A., Srivastava, B., and Sheth, A. (2021). "Who can help me?: Knowledge Infused Matching of Support Seekers and Support Providers during COVID-19 on Reddit." In 2021 IEEE 9th International Conference on Healthcare Informatics (ICHI) (pp. 265-269) [paper]

• Kaushik Roy, Zhang, Q., Gaur, M., & Sheth, A. (2021). Knowledge-infused Policy Gradients with Upper Confidence Bound for Relational Bandits. In Machine Learning and Knowledge Discovery in Databases. Research Track: European Conference, ECML PKDD 2021, Bilbao, Spain, September 13–17, 2021, Proceedings, Springer International Publishing (pp. 35-50) [paper].

• Bhat, A., & Kaushik Roy. (2016, August). Optimized KNN for Sequential Execution. In 2016 International Conference on Inventive Computation Technologies (ICICT) (Vol. 1, pp. 1-6) [paper].

• Nasreen, A., Kaushik Roy, Roy, K., & Shobha, G. (2015, June). Key Frame Extraction and Foreground Modelling using K-means Clustering. In 2015 7th International Conference on Computational Intelligence, Communication Systems and Networks (pp. 141-145) [paper].

• Bhat, A. G., Kaushik Roy, Anchalia, P. P., & Jeevith, H. M. (2015, March). Design and Implementation of a Dynamic Intelligent Traffic Control System. In 2015 17th UKSim-AMSS International Conference on Modelling and Simulation (UKSim) (pp. 369-373) [paper].

• Anchalia, P. P., Kaushik Roy, & Roy, K. (2015, March). Two Class Fisher's Linear Discriminant Analysis Using MapReduce. In 2015 17th UKSim-AMSS International Conference on Modelling and Simulation (UKSim) (pp. 463-469) [paper].

• Anchalia, P. P., & Kaushik Roy (2014, January). The K-nearest Neighbor Algorithm using MapReduce Paradigm. In 2014 5th International conference on intelligent systems, modelling and simulation (pp. 513-518) [paper].

SYMPOSIUM AND WORKSHOP PUBLICATIONS

• Kaushik Roy, Zi, Y., & Sheth, A. (2024, November). Towards Pragmatic Temporal Alignment in Stateful Generative AI Systems: A Configurable Approach. In Proceedings of the AAAI Symposium Series (Vol. 4, No. 1, pp. 388-390) [paper].

• Kaushik Roy, Oltramari, A., Zi, Y., Shyalika, C., Narayanan, V., & Sheth, A. (2024, May). Causal Event Graph-Guided Language-based Spatiotemporal Question Answering. In Proceedings of the AAAI Symposium Series (Vol. 3, No. 1, pp. 227-233) [paper].

• Zi, Y., Kaushik Roy, Narayanan, V., & Sheth, A. (2024, May). Exploring Alternative Approaches to Language Modeling for Learning from Data and Knowledge. In Proceedings of the AAAI Symposium Series (Vol. 3, No. 1, pp. 279-286) [paper].

• Raj, K., Kaushik Roy, Bonagiri, V., Govil, P., Thirunarayan, K., Goswami, R., & Gaur, M. (2024, May). K-PERM: Personalized Response Generation Using Dynamic Knowledge Retrieval and Persona-Adaptive Queries. In Proceedings of the AAAI Symposium Series (Vol. 3, No. 1, pp. 219-226) [paper].

• Zi, Y., Veeramani, H., Kaushik Roy, & Sheth, A. P. (2023, December). RDR: The Recap, Deliberate, and Respond Method for Enhanced Language Understanding. In Neuro-Symbolic Learning and Reasoning in the era of Large Language Models [paper].

• Shiri, A., Kaushik Roy, Sheth, A., & Gaur, M. (2024, January). L3 Ensembles: Lifelong Learning Approach for Ensemble of Foundational Language Models✶. In Proceedings of the 7th Joint International Conference on Data Science & Management of Data (11th ACM IKDD CODS and 29th COMAD) (pp. 592-594) [paper].

• Kaushik Roy, Zi, Y., Gaur, M., Malekar, J., Zhang, Q., Narayanan, V., & Sheth, A. (2023, October). Process Knowledge-infused Learning for Clinician-friendly Explanations. In Proceedings of the AAAI Symposium Series (Vol. 1, No. 1, pp. 154-160) [paper].

• Tsakalidis, A., Chim, J., Bilal, I. M., Zirikly, A., Atzil-Slonim, D., Nanni, F., ..., Kaushik Roy, ... & Liakata, M. (2022, July). Overview of the CLPsych 2022 Shared Task: Capturing Moments of Change in Longitudinal User Posts. In Proceedings of the Eighth Workshop on Computational Linguistics and Clinical Psychology (pp. 184-198) [paper].

• Gupta, S., Agarwal, A., Gaur, M., Kaushik Roy, Narayanan, V., Kumaraguru, P., & Sheth, A. (2022). Learning to Automate Follow-up Question Generation Using Process Knowledge for Depression Triage on Reddit Posts. In Proceedings of the Eighth Workshop on Computational Linguistics and Clinical Psychology (pp. 137-147) [paper].

• Kaushik Roy, Rawte, V., Chakraborty, M., Gaur, M., Faldu, K., Kikani, P., ... & Sheth, A. P. TDLR: Top Semantic-Down Syntactic Language Representation. In NeurIPS'22 Workshop on All Things Attention: Bridging Different Perspectives on Attention [paper].

• Lokala, U., Lamy, F., Dastidar, T. G., Kaushik Roy, Daniulaityte, R., Parthasarathy, S., & Sheth, A. (2021). eDarkTrends: Harnessing Social Media Trends in Substance Use Disorders for Opioid Listings on Cryptomarket. In ICLR AI for Public Health Workshop 2021 [paper].

SELECTED TALKS AND TUTORIALS

• 2024: Neurosymbolic Customized and Compact CoPilots, International Semantic Web Conference tutorial.

• 2024: Healthcare Assistance Challenges-Driven Neurosymbolic AI, Ontolog Summit invited talk.

• 2024: Knowledge-infused Neurosymbolic AI for Mental Healthcare, University of Florida ICCC.

• 2023: Causal AI for Web and Health Care, ACM Web Conference tutorial.

• 2022: Knowledge-infused Reinforcement Learning, Knowledge Graph Conference tutorial.

• 2020: Knowledge-infused Statistical Learning for Social Good Applications, PyData Berlin.

SELECTED PROFESSIONAL SERVICE

• Guest Editor, IEEE Internet Computing special issue on Civilizing and Humanizing AI.

• Guest Editor, Information special issue on information integration, language models, and knowledge graphs.

• Program Committee, AAAI 2024 Fall Symposium on Large Language Models for Knowledge Graph and Ontology Engineering.

• Organizing Committee, KiL 2023 at KDD.

• Organizing Committee, Collaborative Assistants for the Society (CASY), 2022.

MENTORING AND TEACHING

• Doctoral student supervision: Ashraf Abdul-Muumin, University of Alabama.

• Curriculum development: Computational Foundations of Machine Learning (CS 483/583); AI and MLOps topics.

MEDIA AND OUTREACH

• AI in the Family (2025) — film listed on IMDb, https://www.imdb.com/title/tt38192436/ (specific credit should be verified).

• Recorded public talks and tutorials on YouTube; see faculty website's Beyond Research page.

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